

Hostel- Hunting using Fuzzy Logic

Internal Working of fuzzy_engine.py

Overview

fuzzy_engine.py is the core component of the Hostel Hunter. It implements the fuzzy logic recommendation engine responsible for evaluating hostels and generating personalized rankings based on user preferences. The engine follows four main stages:

1. Dataset Loading
2. Fuzzification
3. Fuzzy Scoring & Rule Evaluation
4. Ranking & Recommendation

1.Dataset Loading

When the Fuzzy Recommender class is initialized, the system loads the hostel dataset from: data/bern_hostel_complete_dataset.csv

The dataset is converted into a list of hostel records for efficient processing.

Loaded Attributes

- Hostel Name
- Area
- Budget per Night
- Distance from Centre
- Safety Rating
- Cleanliness Rating
- Social Vibe Score
- Wi-Fi Quality
- Breakfast Availability
- Station Availability

2. Traveler Profiles

The system supports three traveller categories.

Traveller Type	Backpacker	Social Butterfly	Worry-Free
Prioritize	• Affordability • Experience • Convenience	• Social atmosphere • Community experience	• Safety • Cleanliness • Reliable accommodation
Weights	Value = 30% Convenience = 20% Experience = 28% Safety = 22%	Value = 18% Convenience = 18% Experience = 46% Safety = 18%	Value = 15% Convenience = 15% Experience = 15% Safety = 55%

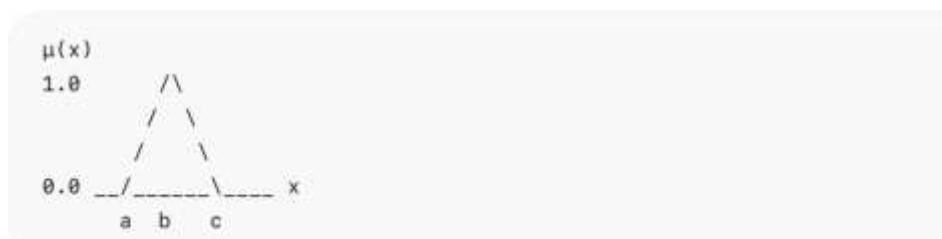
3. Membership Functions

The fuzzy engine uses two membership functions:

Triangular Membership Function- Used for medium fuzzy sets.

$$\mu(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ \frac{c-x}{c-b}, & b < x < c \\ 0, & x \geq c \end{cases}$$

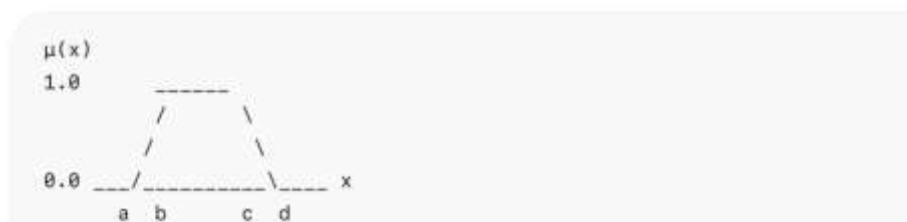
Graph:



Trapezoidal Membership Function - Used for low and high fuzzy sets.

$$\mu(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ 1, & b < x \leq c \\ \frac{d-x}{d-c}, & c < x < d \\ 0, & x \geq d \end{cases}$$

Graph:



These functions convert numerical values into fuzzy linguistic values.

4. Fuzzification

Fuzzification transforms crisp hostel attributes into fuzzy sets.

Price Fuzzification

Example: Price = €30

Low = 0.67, Medium = 0.25, High = 0

Distance Fuzzification

Example: Distance = 2 km

Near = 0.80 Moderate = 0.20 Far = 0

Rating Fuzzification

Applied to: Safety, Cleanliness, Social Vibe, Wi-Fi

Categories: Low Medium High

Ratings are converted into membership values between 0 and 1.

5.COMPONENT SCORING & FUZZY RULE ENGINE

a. Value Score Calculation:

The Value Module evaluates hostel affordability.



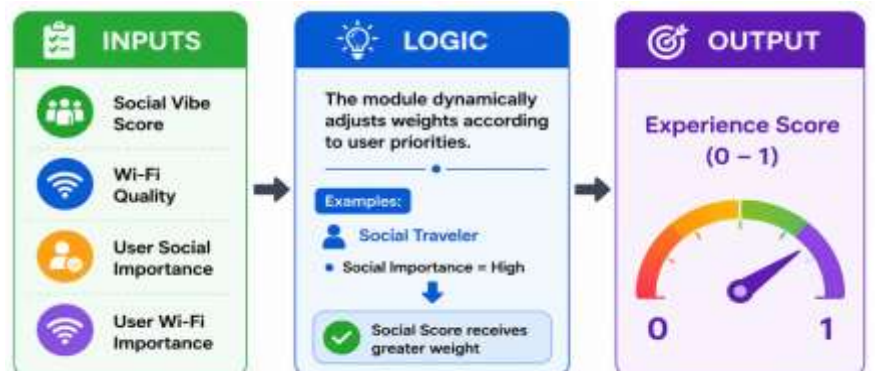
b. Convenience Score Calculation:

The Convenience Module evaluates accessibility.



c. Experience Score Calculation:

The Experience Module evaluates comfort and enjoyment.



d. Safety Score Calculation:

The Safety Module evaluates security and cleanliness.



6. Fuzzy Rule Evaluation

The system contains 10 fuzzy inference rules.

- IF price is **LOW** AND safety is **HIGH** → recommendation **HIGH**
- IF price is **LOW** AND safety is **MEDIUM** → recommendation **VERY GOOD**
- IF distance is **FAR** AND price is **HIGH** → recommendation **LOW**
- IF safety is **LOW** → recommendation **VERY LOW**
- IF cleanliness is **HIGH** AND safety is **HIGH** → recommendation **HIGH**
- IF social vibe is **HIGH** AND price is **LOW** → recommendation **HIGH**
- IF distance is **NEAR** AND price is **LOW** → recommendation **HIGH**
- IF distance is **NEAR** AND safety is **HIGH** → recommendation **VERY GOOD**
- IF price is **MEDIUM** AND cleanliness is **HIGH** → recommendation **GOOD**
- IF social vibe is **HIGH** AND cleanliness is **HIGH** → recommendation **VERY GOOD**

Each rule produces a recommendation strength value.

7. Defuzzification

The engine uses the **Centre of Gravity (COG)** method. The weighted average of all activated rules produces a crisp recommendation value.

Output range:

0.0 → Poor Match

1.0 → Excellent Match

8. Final Score Calculation

Two independent scores are generated:

Layered Weighted Score

Combines: Value , Convenience, Experience, Safety. Using traveller profile weights.

Rule-Based Fuzzy Score

Generated from fuzzy inference rules. Final formula:

$$\text{Final Recommendation Score} = (\text{Weighted Score} \times 0.60) + (\text{Fuzzy Rule Score} \times 0.40)$$

This balances direct preference matching and fuzzy reasoning.

9. Recommendation Filtering

Before scoring, hostels are filtered according to:

- Minimum Safety Requirement
- Minimum Cleanliness Requirement
- Maximum Distance
- Breakfast Requirement
- Station Requirement
- Hostels failing these conditions are removed.

10. Ranking Process

After scoring:

1. Every hostel receives a final recommendation score.
2. Hostels are sorted in descending order.
3. Highest-scoring hostels appear first.
4. On Clicking in the hostel , it shows below info:



Summary

fuzzy_engine.py acts as the intelligent decision-making module of the system. It converts hostel attributes into fuzzy values, evaluates them through four specialized modules (Value, Convenience, Experience, and Safety), applies fuzzy inference rules, performs defuzzification, and finally ranks hostels according to how well they match the user's preferences. This approach allows the system to mimic human decision-making and provide personalized hostel recommendations.